Optimization Methods for Machine Learning - Fall 2025

PROJECT # 2
Support Vector Machines

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Instructions

The project will be developed in team, the same as the first project unless severe exceptions: each team must hand in their own answers. We will be assuming that, as participants in a graduate course, every single student will be taking the responsibility to make sure his/her personal understanding of the solution to any work arising from such collaboration.

Submission

Projects must be uploaded using a google form

https://forms.gle/3xxKGR6CXUgK4qpW8

Each team leader is in charge of uploading the files.

The team must upload the python files organized as requested in the instructions at the end of the project, in a compressed folder named "GroupNumber"_"GroupName"_code.zip.
A typed report in English and in pdf format must be uploaded too. **The report must be of at most 4 pages, excluded figures that must be put at the end. Do not include part of the python code.** The report must be named "GroupNumber"_"GroupName"_report.pdf

Evaluation criteria

The grade follows the Italian system, with scores in the range [0,31], being 18 the minimum degree to pass the exam and 31 corresponding to 30 cum laude.
The project is due at the latest at midnight on the due date. For late submissions, the score will be decreased. It is worth 85% for the next 48 hours. It is worth 70% from 48 to 120 hours after the due date. It is worth 50% credit after a 120-hour delay.
The second project accounts for 30% of the total vote of the exam.
For the evaluation of the second project, the following criteria will be used:

1. 60% check of the implementation
2. 40% quality of the overall project as explained in the report.

The grade of the first project will be assigned AFTER a **project revision** with teacher and assistant of the submitted code, during which each student **must** be able to explain their own work and contributions. Tentative date for the project revision is December 22 in the morning.

Assignment Description

In this project, you will implement different optimization methods for training Support Vector Machines (supervised learning) in order to solve a classification problem.

Datasets.
You are asked to build a classifier that distinguishes between images of Fashion MNIST dataset. Each sample is a 28x28 grayscale image, associated with a label.
It is necessary to scale the data before the optimization.

You will be given a full dataset and you will extract the target set made up of images belonging to two (Question 1,2,3) or three (Question 4) classes. Instructions on how to import the data in Python will be provided in a separate file. Train and test datasets are already provided as np.array objects in the extractor.py file. For Questions 1, 2 and 3, the target set will be made of images of pullovers and dresses (identified by the labels 2 and 3, in the target set, respectively). In Question 4, you will have to extract the images of shirts with label 6. A $k-$fold cross-validation can be used to set the hyperparameters. Please note that the teacher has defined a test set that might be used to check the accuracy of your models.

The SVM problem Consider a nonlinear SVM with kernel $k(\cdot,\cdot)$ and the corresponding nonlinear decision function

$$y(x) = sign\left(\sum_{p=1}^{P} \alpha_p y^p k(x^p, x) + b\right)$$

where α, b are obtained as the optimal solution of the dual nonlinear SVM problem.

As kernel function you may choose either a RBF kernel

$$k(x,y) = e^{-\gamma\|x-y\|^2}$$

where $\gamma > 0$ is an hyper parameter, or a polynomial kernel

$$k(x,y) = (x^T y + 1)^\gamma$$

with hyper parameter $\gamma \geq 1$.

You are requested to train the SVM, namely to both set the values of the hyperparameters C and γ with a heuristic procedure, and to find the values of the parameters α, b with an optimization procedure. For this last task, you need to follow the different optimization procedures described in the following questions.

Question 1. (max score up to 23) Write a program to find the solution of the SVM dual quadratic problem. Perform a grid search with k-fold cross validation to set the values of the hyperparameters C and γ. To solve the SVM dual quadratic problem, you can use any routine that relies on the gradient of the objective function, which can be easily computed. Note that using a *specific* method for Quadratic Programming problems would be preferable and it may lead to higher grade.

Question 2. (up to "+4" points) Use the same kernel and hyperparameters of Question 1. Write a program which implements a decomposition method for the dual quadratic problem with any **even** value $\mathbf{q} \geq \mathbf{4}$. You must define the selection rule of the working set, construct the subproblem at each iteration and use a standard algorithm for its solution. You can use any routine that exploits the gradient of the objective function (which is simply computable). However, the use of the specific method for quadratic programming would be preferable and it may lead to higher grade. You must implement a stopping criterion in the outer optimization loop based on the optimality conditions.

Question 3. (up to "+3" points) Use the same kernel and hyperparameters of Question 1. Fix the dimension of the sub-problem to $q = 2$ and implement the most violating pair (MVP) decomposition method, which uses the analytic solution of the sub-problems. You must implement a stopping criterion in the outer optimization loop based on the optimality conditions.

Question 4. (up to "+1" points)
Consider a three-class problem with the classes of pullovers, dresses, and shirts. Implement an SVM strategy for multiclass classification (either one against one or one against all). You are free to use any method you find most suitable for solving the dual SVM problems.

Instructions for the report

In the report you must state:

- The chosen kernel;
- Only for Question 1: the final setting for the hyperparameter C (upper bound of the constraints of the dual problem) and of the hyperparameter of the kernel chosen; how these values were selected and whether you could identify values that highlight over/under fitting;
- Only for Question 1 and Question 2: which optimization routine you used for solving the quadratic minimization problem and the setting of its parameters (if any);
- For each Question:
    - Machine learning performance: report the values of the accuracy on the training and test sets; (for Question 1, also report the value of the validation accuracy, if computed).

- Optimization performance: report the initial and final value of the objective function of the dual problem, the number of iterations, the number of function evaluations, and the violation of the KKT condition (either as it is returned by the optimization routine or evaluated by yourselves).

The comparison among all the implemented methods - in terms of accuracy in learning and computational effort in training - must be gathered into a final table (an example below).

	hyperparameters			ML performance		optimization performance			
question	C	γ	q	Training accuracy	Test accuracy	number its	number fun evals	KKT viol	cpu time
Q1									
Q2	As in Q1								
Q3	As in Q1								
Q4									

Instructions for python code

In the folder you submit, you **must** include the folder *FashionMNIST* containing the training and test CSV datasets, which will be used by your scripts to complete the tasks.
For each question $i = \{1,2,3,4\}$ you have to create a different folder named **Q_i** where **you must provide the files according to the following instructions:**

- A file called **run_i_GroupName.py** (e.g. run_1_Example.py,). This file will be the only one executed in the phase of verification of the work done. It can include all the classes, and import all the functions and libraries you used for solving the specific question i, but it has to print ONLY:
  - Setting values of the hyperparameters
  - Classification rate on the training set (% instances correctly classified)
  - Classification rate on the test set (% instances correctly classified)
  - The confusion matrix
  - Time necessary for the optimization
  - Number of optimization iterations
  - Difference between $m(\alpha)$ and $M(\alpha)$
  - The final value of the Dual SVM objective function $f(\alpha^*)$
  - Solver status (e.g. optimal, infeasible etc..)

  Furthermore, in Question 2 you must print also:

  - Value of q

In addition, in Question 3 you must print also:

- Number of columns of the matrix in the quadratic objective function (i.e. Q) computed to solve the dual SVM.

Finally, in Question 4 you must print also:

- if you used OAA or OAO (i.e., the multiclass strategy that you implemented).

- A file called **functions_i_GroupName.py** with the complete code you have written that includes all the functions that are called from the **run_i_GroupName.py file** and all the procedures that have been used to select the hyperparameters.

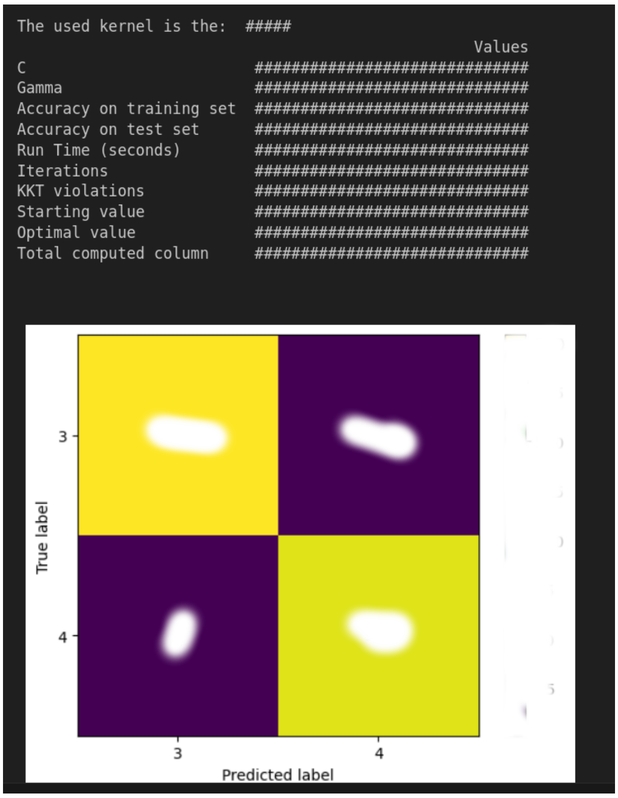

Figure 1: Example of the expected output of file run_i_GroupName.py

Note that:

1. A test error on a blind test set might be evaluated to check the performance of the models.
2. There is no competition with other students. Your grade does not depend on other teams' performances.

3. You are warmly recommended to write clean and easily readable code. Not respecting the general assignment (for example, printing different values or not printing the requested outputs) will result in a penalization. Printing the progress of the algorithm or similar mistakes will slow down the code correction and result in a score decrease. Additionally, please make sure to comment your code sufficiently, so that an external reader can understand the purpose of each part of the code.

4. Your code **will not be debugged** by the teachers and it will not be possible to modify it after the submission. You are strongly recommended to check multiple times and on different computers that your code is executable.

5. It is welcomed to take inspiration from other students, as well as from open-source contents online, but **"copy and paste" strategies will invalidate your project**. Be aware that **your code will be cross-checked with online materials, as well as with other students' code and with AI-generated code**.

6. **Responsible use of AI.**
   A responsible use of AI tools (e.g., ChatGPT, Copilot, etc.) is allowed, but should only be used as a **support** for the implementation process (e.g., explore function usage, assist with debugging). Students **must** always verify, fully understand, and adapt any AI-assisted output, ensuring complete awareness and control of their code and methodological choices. Using AI-generated code without proper understanding and adaptation will be considered inappropriate and will result in a penalization of the score during the discussion phase.
   In case you used AI tools during the development of your project, you **must** state in the report how you used them and for what purpose.

7. The use of any library automatically building ML models (e.g, Pytorch, TensorFlow, sklearn.svm etc.) is **FORBIDDEN** and automatically invalidates the project.